

Calibration results

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Normalized Residuals

Reprojection error (cam0): mean 0.4709523407464676, median 0.3991977233618694, std: 0.3372425317661671
Gyroscope error (imu0): mean 3.305496916810419, median 2.3274982163081406, std: 5.677777796030888
Accelerometer error (imu0): mean 5.942728742680633, median 3.7275797717127346, std: 13.595125216661023

Residuals

Reprojection error (cam0) [px]: mean 0.4709523407464676, median 0.3991977233618694, std: 0.3372425317661671
Gyroscope error (imu0) [rad/s]: mean 0.001896173463673923, median 0.0013351518593369188, std: 0.0032570145610237903
Accelerometer error (imu0) [m/s²]: mean 0.042105064799677114, median 0.026410424340373585, std: 0.09632336473581479

Transformation (cam0):

T_ci: (imu0 to cam0):

```
[[-0.01220395  0.99926453  0.03635189  0.04974994]
 [ 0.03002707  0.03670443 -0.99887495  0.05003318]
 [-0.99947458 -0.01109867 -0.03045292 -0.11566127]
 [ 0.          0.          0.          1.          ]]
```

T_ic: (cam0 to imu0):

```
[[-0.01220395  0.03002707 -0.99947458 -0.1164957 ]
 [ 0.99926453  0.03670443 -0.01109867 -0.05283348]
 [ 0.03635189 -0.99887495 -0.03045292  0.04464616]
 [ 0.          0.          0.          1.          ]]
```

timeshift cam0 to imu0: [s] ($t_{imu} = t_{cam} + \text{shift}$)
-0.08532874252196998

Gravity vector in target coords: [m/s²]

```
[ 0.03090526  1.55083938 -9.68309687]
```

Calibration configuration

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cam0

Camera model: pinhole

Focal length: [158.65126400023672, 159.25587984496482]

Principal point: [160.16619284096896, 159.21246044902614]

Distortion model: radtan

Distortion coefficients: [-0.0032825513758021696, -6.574744393870335e-05, -0.00013568989651293803, -0.000884999262477561]

Type: checkerboard

Rows

Count: 8

Distance: 0.02 [m]

Cols

Count: 11

Distance: 0.02 [m]

IMU configuration

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IMU0:

Model: calibrated

Update rate: 50.0

Accelerometer:

Noise density: 0.0010019901002151698

Noise density (discrete): 0.007085139945439349

Random walk: 1.9893785362909353e-05

Gyroscope:

Noise density: 8.112529814510266e-05

Noise density (discrete): 0.0005736424844418254

Random walk: 6.28721982800146e-07

T_ib (imu0 to imu0)

[[1. 0. 0. 0.]

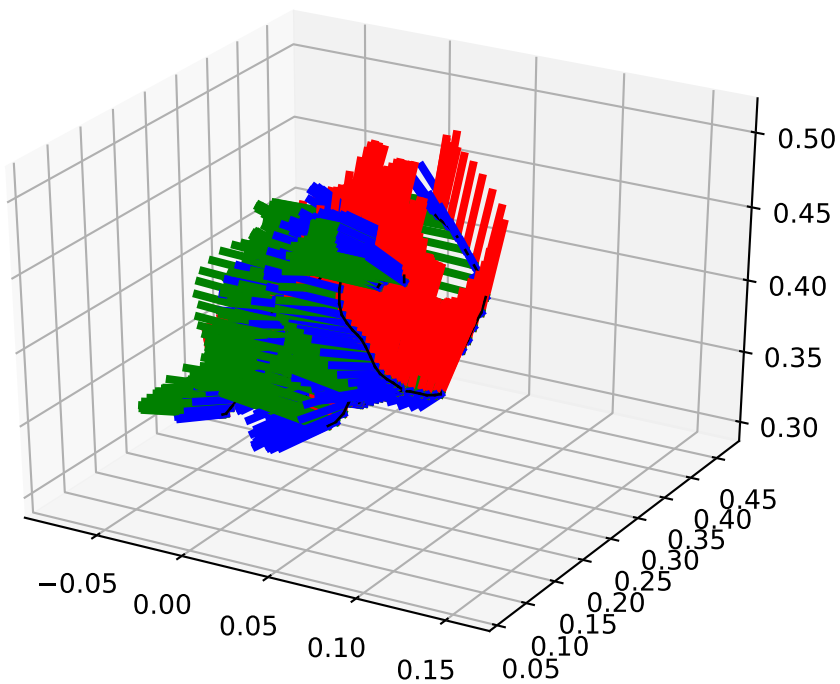
[0. 1. 0. 0.]

[0. 0. 1. 0.]

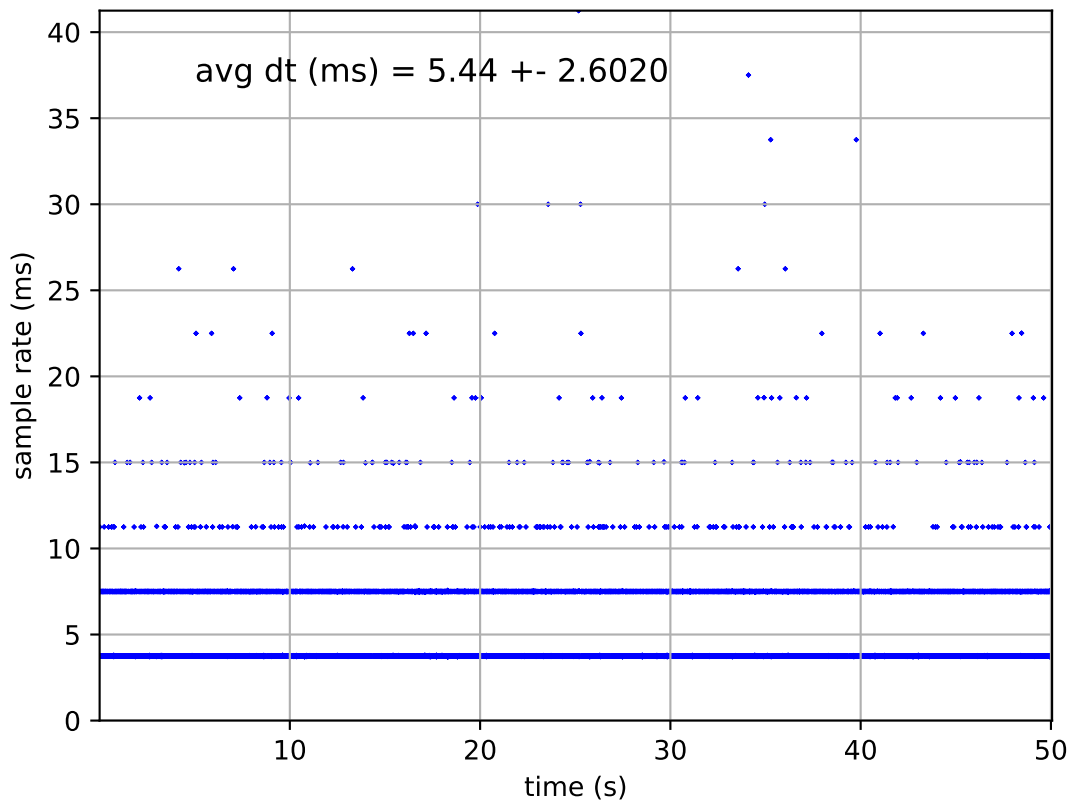
[0. 0. 0. 1.]]

time offset with respect to IMU0: 0.0 [s]

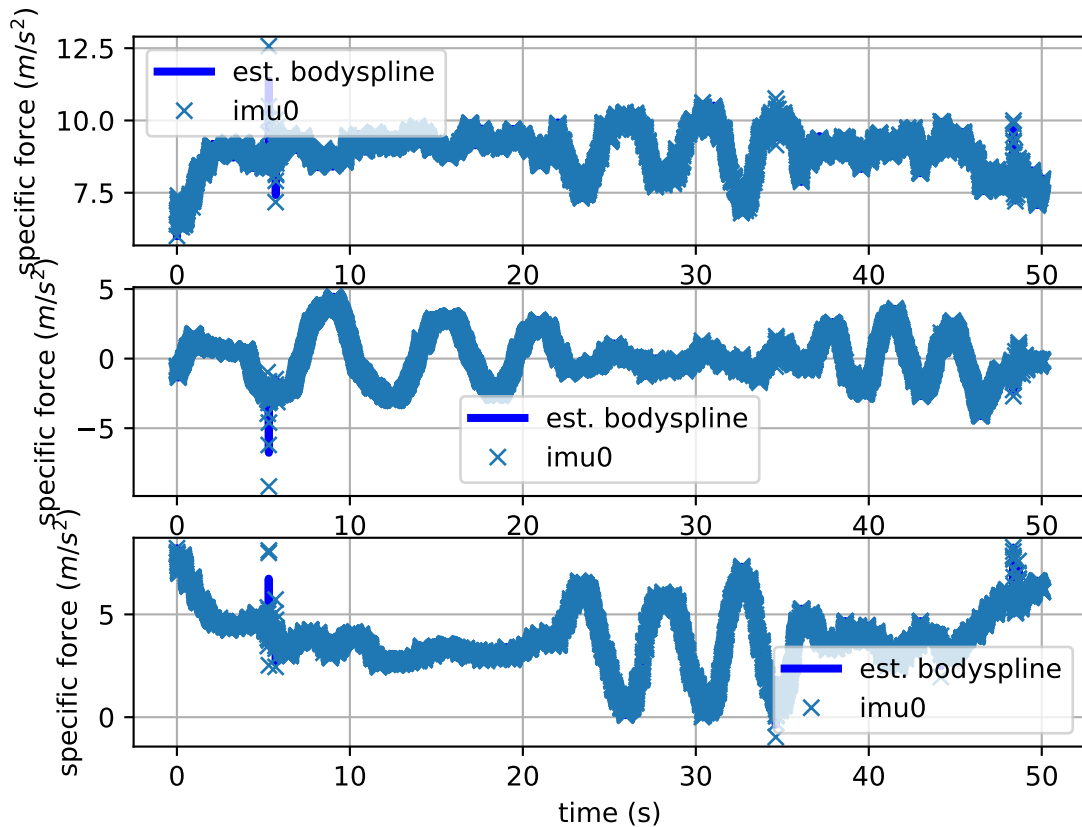
imu0: estimated poses



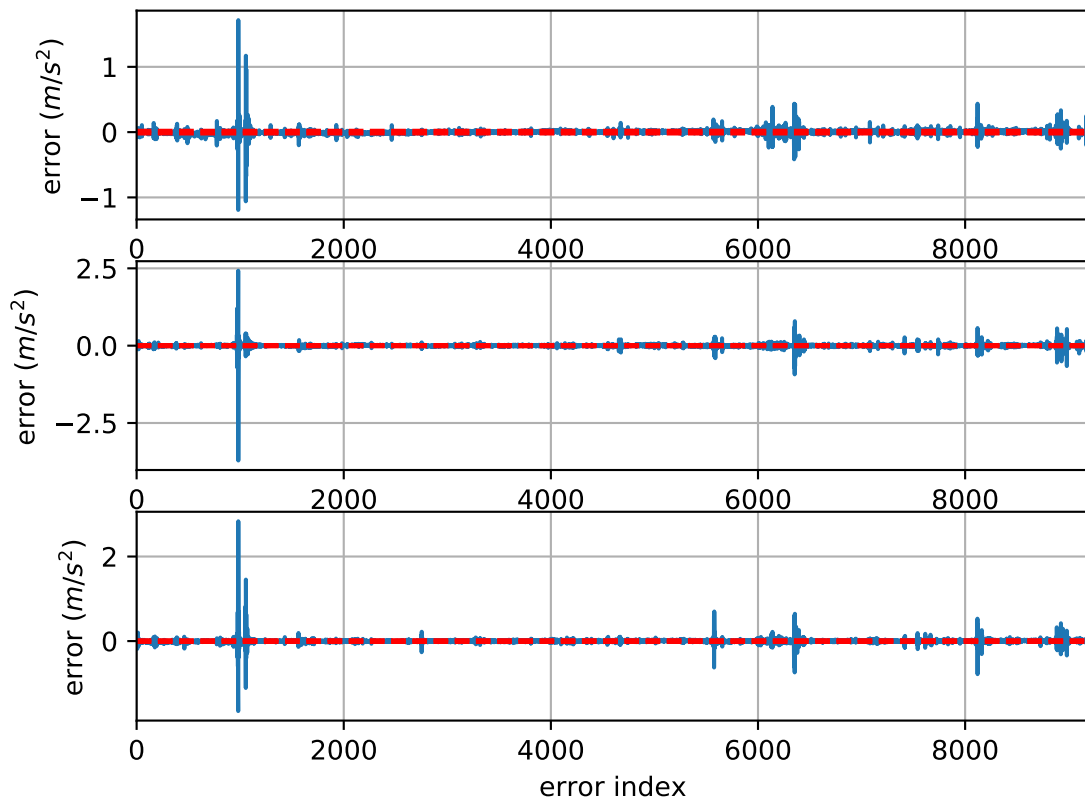
imu0: sample inertial rate



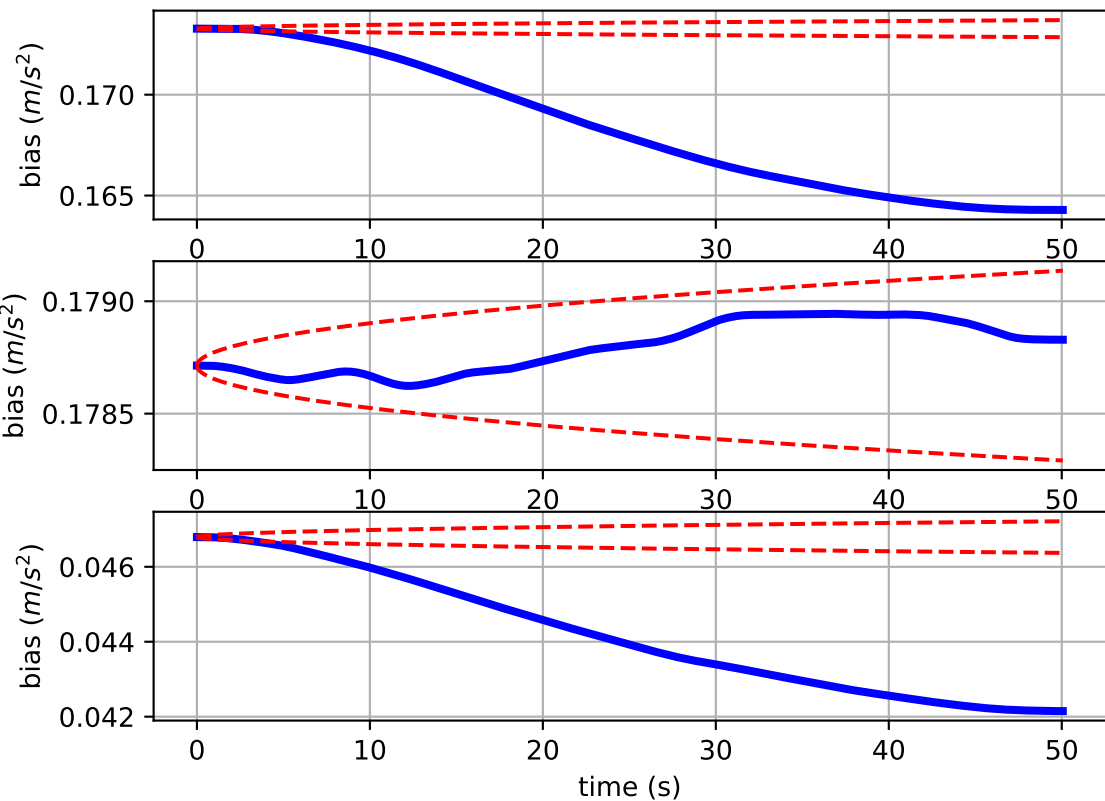
Comparison of predicted and measured specific force (imu0 frame)



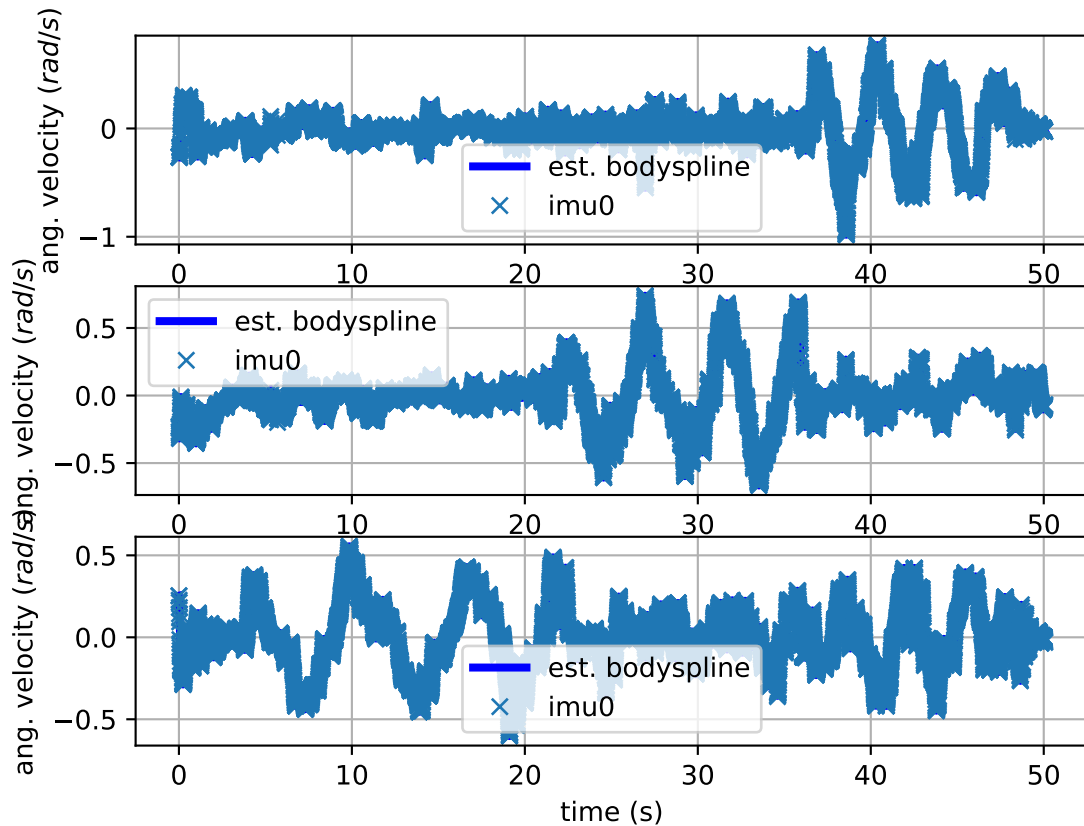
imu0: acceleration error



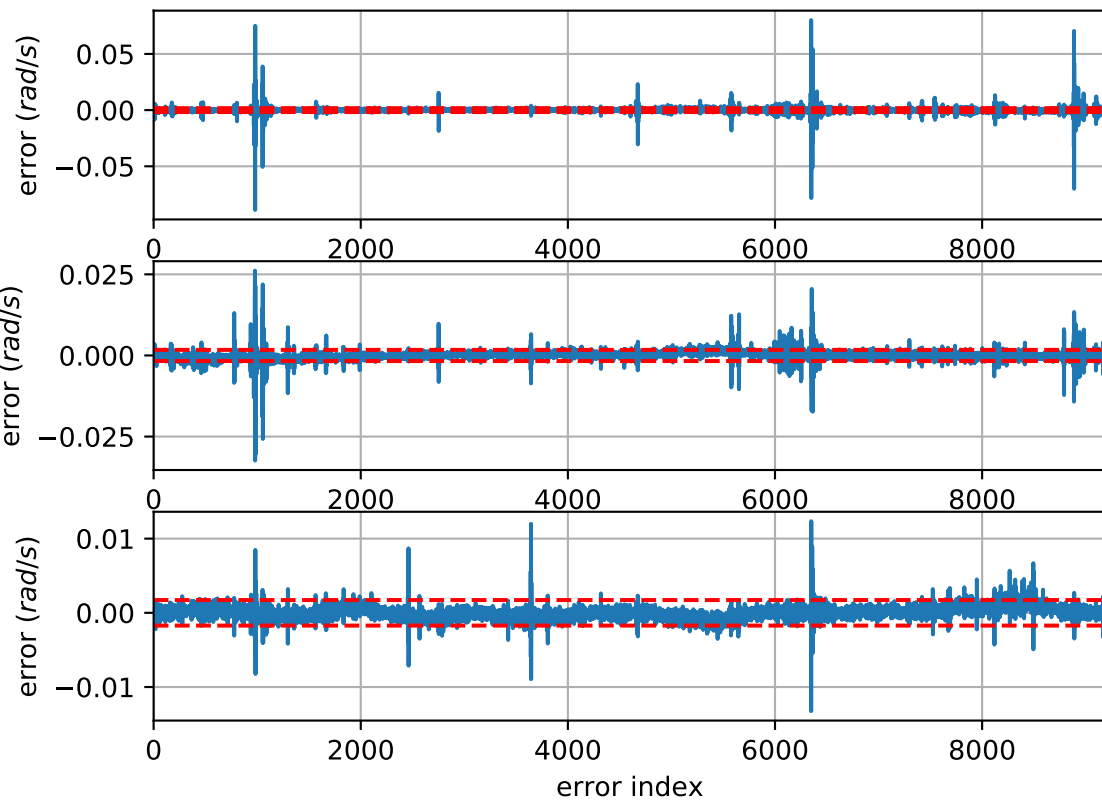
imu0: estimated accelerometer bias (imu frame)



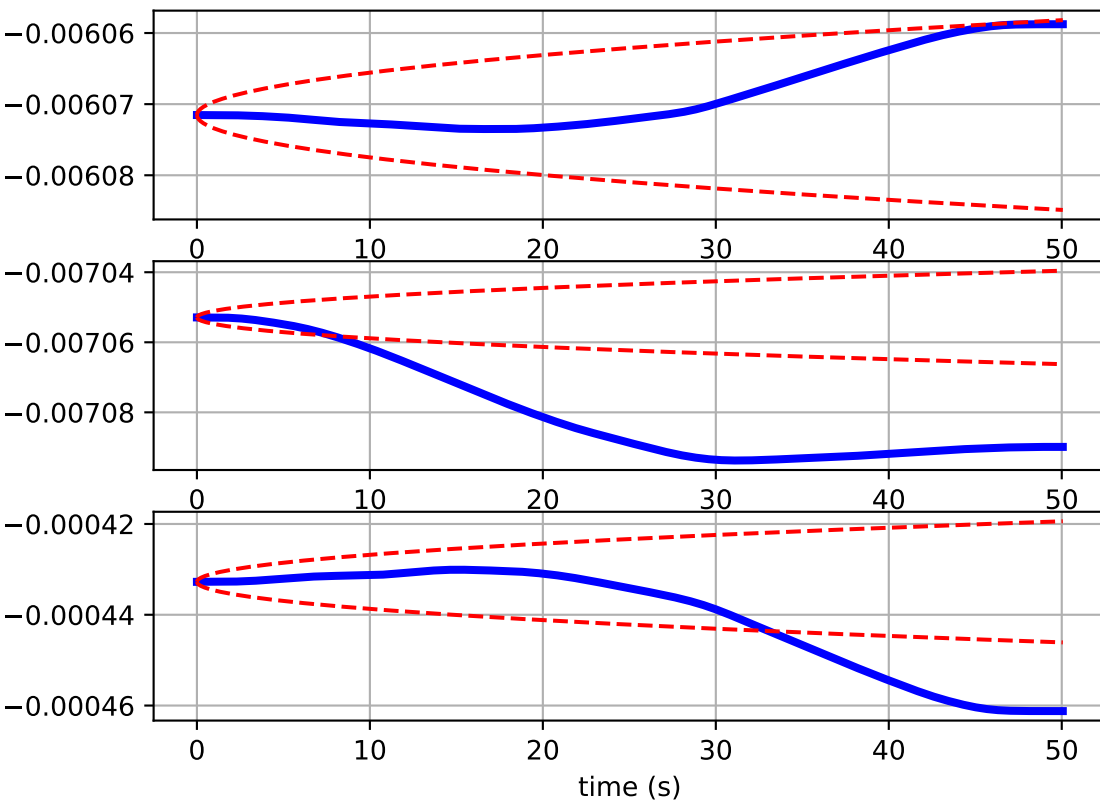
Comparison of predicted and measured angular velocities (body frame)



imu0: angular velocities error



imu0: estimated gyro bias (imu frame)



cam0: reprojection errors

